Solutions to the Exercises of Chapter 5 (Linear Systems)

Strogatz, *Nonlinear Dynamics and Chaos* (2nd ed.)

Conventions. For $\dot{\mathbf{x}} = A\mathbf{x}$ with A a real 2×2 matrix, $\tau = \operatorname{tr} A$, $\Delta = \det A$, and the eigenvalues are $\lambda_{1,2} = \frac{1}{2}\big(\tau \pm \sqrt{\tau^2 - 4\Delta}\big)$. I classify the origin with the usual trace–determinant rules: $\Delta < 0$ saddle; $\Delta > 0$ and $\tau^2 > 4\Delta$ node (stable if $\tau < 0$, unstable if $\tau > 0$); $\Delta > 0$, $\tau^2 < 4\Delta$ spiral ($\tau \neq 0$) or center ($\tau = 0$); $\tau^2 = 4\Delta$ star or degenerate node; $\Delta = 0$ non-isolated fixed points.

Phase portraits drawn in this document were generated from the exact solution $\mathbf{x}(t) = e^{At}\mathbf{x}_0$ (blue curves, arrows give the direction of motion); dashed red lines are real eigendirections and a thick red line marks a line of fixed points. Where a problem asks for a sketch of the vector field or of trajectories, I also describe the picture in words.

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1 Definitions and Examples (5.1)

5.1.1 *(Ellipses and energy conservation for the harmonic oscillator.)* $\dot{x} = v$, $\dot{v} = -\omega^2 x$. *(a) Show the orbits are the ellipses* $\omega^2 x^2 + v^2 = C$. *(b) Show this is equivalent to conservation of energy.*

Solution. *(a)* Dividing the equations, $\dfrac{\mathrm{d}v}{\mathrm{d}x} = \dfrac{\dot{v}}{\dot{x}} = -\dfrac{\omega^2 x}{v}$, i.e. $v\,\mathrm{d}v = -\omega^2 x\,\mathrm{d}x$. Integrating, $\frac{1}{2}v^2 = -\frac{1}{2}\omega^2 x^2 + \text{const}$, so

$$\omega^2 x^2 + v^2 = C, \qquad C \geq 0.$$

For $C > 0$ this is an ellipse with semi-axes $\sqrt{C}/\omega$ (in x) and $\sqrt{C}$ (in v); $C = 0$ is the origin.

(b) The system is $\ddot{x} = -\omega^2 x$, i.e. a unit mass on a spring with constant $k = \omega^2$. Its energy is $E = \frac{1}{2}v^2 + \frac{1}{2}\omega^2 x^2 = \frac{1}{2}C$. Directly, $\dot{E} = v\dot{v} + \omega^2 x\dot{x} = -\omega^2 xv + \omega^2 xv = 0$. So the condition $\omega^2 x^2 + v^2 = C$ is just $E = \text{const}$, conservation of energy. ■

5.1.2 $\dot{x} = ax$, $\dot{y} = -y$ *with* $a < -1$. *Show that all trajectories become parallel to the* y*-direction as* $t \to \infty$ *and to the* x*-direction as* $t \to -\infty$.

Solution. The solution is $x = x_0 e^{at}$, $y = y_0 e^{-t}$ (take $x_0, y_0 \neq 0$; the axes are trivial). The slope is

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\dot{y}}{\dot{x}} = \frac{-y}{ax}, \qquad \left|\frac{y}{x}\right| = \left|\frac{y_0}{x_0}\right| e^{-(1+a)t} = \left|\frac{y_0}{x_0}\right| e^{(|a|-1)t}.$$

Since $a < -1$, $|a| - 1 > 0$. As $t \to +\infty$, $|y/x| \to \infty$, so $|\mathrm{d}y/\mathrm{d}x| = |y/(ax)| \to \infty$: the trajectories become vertical (parallel to the y-direction). As $t \to -\infty$, $|y/x| \to 0$, so $\mathrm{d}y/\mathrm{d}x \to 0$: the trajectories become horizontal. (Here x decays faster than y because $|a| > 1$, so y is the slow direction at $t \to \infty$; backward in time x blows up faster.) ■

5.1.3–5.1.6 *Write the systems in matrix form: (3)* $\dot{x} = -y,\ \dot{y} = -x$*; (4)* $\dot{x} = 3x - 2y,\ \dot{y} = 2y - x$*; (5)* $\dot{x} = 0,\ \dot{y} = x + y$*; (6)* $\dot{x} = x,\ \dot{y} = 5x + y$.

Solution.

5.1.3: $\dot{\mathbf{x}} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}\mathbf{x}$, **5.1.4:** $\dot{\mathbf{x}} = \begin{pmatrix} 3 & -2 \\ -1 & 2 \end{pmatrix}\mathbf{x}$, **5.1.5:** $\dot{\mathbf{x}} = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}\mathbf{x}$, **5.1.6:** $\dot{\mathbf{x}} = \begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix}\mathbf{x}$,

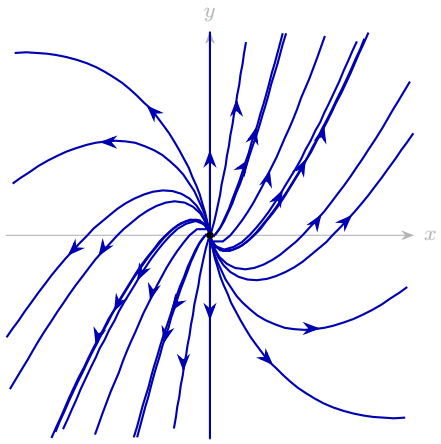
with $\mathbf{x} = (x, y)^T$. ■

5.1.7 *Sketch the vector field and some trajectories of* $\dot{x} = x$, $\dot{y} = x + y$.

Solution. The matrix is $A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. The vector at (x, y) is $(x,\ x + y)$:

point	$(1,0)$	$(0,1)$	$(-1,0)$	$(0,-1)$	$(1,1)$	$(1,-1)$	$(-1,1)$
vector	$(1,1)$	$(0,1)$	$(-1,-1)$	$(0,-1)$	$(1,2)$	$(1,0)$	$(-1,0)$

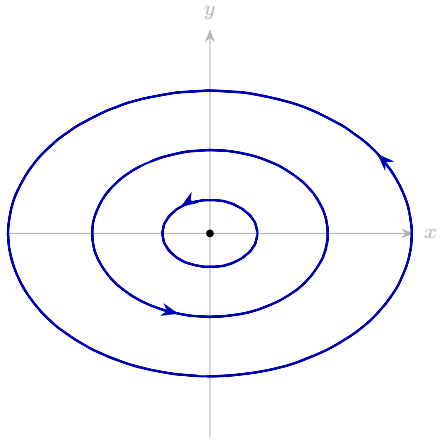
The length of the vectors grows linearly with distance from the origin (the field is linear), and the origin is the only fixed point. The eigenvalue is $\lambda = 1$ (double), with the single eigenvector $(0, 1)$: the origin is a *degenerate unstable node*. Solving, $x = x_0 e^t$, $y = (y_0 + x_0 t)e^t$. The y-axis ($x = 0$) is invariant with $\dot{y} = y$ (outward). Every other trajectory leaves the origin tangent to the y-axis (since $y/x = y_0/x_0 + t \to \pm\infty$ as $t \to -\infty$) and also becomes vertical for large t, so the trajectories bend through an "S" shape.



■

5.1.8 *Sketch the vector field and some trajectories of* $\dot{x} = -2y$, $\dot{y} = x$.

Solution. $A = \begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix}$ has $\tau = 0$, $\Delta = 2$: eigenvalues $\pm i\sqrt{2}$, a *center*. The vector at (x, y) is $(-2y, x)$: at $(1, 0)$ it is $(0, 1)$; at $(0, 1)$ it is $(-2, 0)$; at $(-1, 0)$ it is $(0, -1)$; at $(0, -1)$ it is $(2, 0)$. So the motion is counterclockwise. Because $\frac{\mathrm{d}}{\mathrm{d}t}(x^2 + 2y^2) = 2x(-2y) + 4y(x) = 0$, trajectories are the ellipses $x^2 + 2y^2 = C$, elongated along the x-axis, with period $2\pi/\sqrt{2} = \sqrt{2}\pi$.



■

5.1.9 *Consider* $\dot{x} = -y$, $\dot{y} = -x$. *(a) Sketch the vector field. (b) Show the trajectories are hyperbolas* $x^2 - y^2 = C$. *(c) Find the stable and unstable manifolds of the saddle. (d) Decouple with* $u = x + y$, $v = x - y$ *and solve. (e) Give the manifolds in terms of* u, v. *(f) Give* $x(t), y(t)$.

Solution. *(a)* The vector at (x, y) is $(-y, -x)$: $(1, 0) \mapsto (0, -1)$, $(0, 1) \mapsto (-1, 0)$, $(1, 1) \mapsto (-1, -1)$ (toward the origin along $y = x$), $(1, -1) \mapsto (1, -1)$ (away from the origin along $y = -x$), and so on.

(b) $x\dot{x} - y\dot{y} = x(-y) - y(-x) = 0$, so $\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}(x^2 - y^2)\right) = 0$ and $x^2 - y^2 = C$: hyperbolas (the lines $y = \pm x$ for $C = 0$).

(c) $A = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ has eigenvalues ± 1: for $\lambda = -1$, $(A + I)\mathbf{v} = 0$ gives $\mathbf{v} = (1, 1)$; for $\lambda = +1$, $(A - I)\mathbf{v} = 0$ gives $\mathbf{v} = (1, -1)$. So the origin is a saddle with

$$\text{stable manifold: } y = x, \qquad \text{unstable manifold: } y = -x.$$

(d) $\dot u = \dot x + \dot y = -y - x = -u$ and $\dot v = \dot x - \dot y = -y + x = v$. Hence

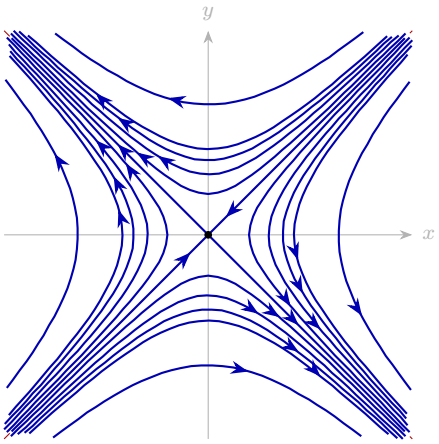
$$u(t) = u_0 e^{-t}, \qquad v(t) = v_0 e^{t}, \qquad u_0 = x_0 + y_0,\ v_0 = x_0 - y_0.$$

(e) Points on the stable manifold $y = x$ have $v = 0$ (they decay, $u \to 0$): the stable manifold is the u-axis $\{v = 0\}$. The unstable manifold $y = -x$ is $\{u = 0\}$, the v-axis.

(f) $x = \frac{1}{2}(u + v)$, $y = \frac{1}{2}(u - v)$, so

$$x(t) = \tfrac{1}{2}\left[(x_0 + y_0)e^{-t} + (x_0 - y_0)e^{t}\right] = x_0 \cosh t - y_0 \sinh t,$$

$$y(t) = \tfrac{1}{2}\left[(x_0 + y_0)e^{-t} - (x_0 - y_0)e^{t}\right] = y_0 \cosh t - x_0 \sinh t.$$



■

5.1.10 *For each system decide whether the origin is attracting, Liapunov stable, asymptotically stable, or none of these: (a) $\dot x = y, \dot y = -4x$; (b) $\dot x = 2y, \dot y = x$; (c) $\dot x = 0, \dot y = x$; (d) $\dot x = 0, \dot y = -y$; (e) $\dot x = -x, \dot y = -5y$; (f) $\dot x = x, \dot y = y$.*

Solution.

(a)	center ($\lambda = \pm 2i$)	Liapunov stable, *not* attracting
(b)	saddle ($\lambda = \pm\sqrt{2}$)	none of the above
(c)	$x = x_0$, $y = y_0 + x_0 t$ (shear)	none of the above
(d)	$x = x_0$, $y = y_0 e^{-t}$	Liapunov stable, *not* attracting
(e)	stable node	asymptotically stable (attracting and Liapunov stable)
(f)	unstable star	none of the above

Proofs are in 5.1.11. ■

5.1.11 *Prove your answers to 5.1.10, producing a suitable δ (attracting) or $\delta(\varepsilon)$ (Liapunov stability).*

Solution. Norms are Euclidean and $\mathbf{x}_0 = (x_0, y_0)$.

(a) $x(t) = x_0 \cos 2t + \frac{1}{2} y_0 \sin 2t$, $y(t) = y_0 \cos 2t - 2x_0 \sin 2t$, and $4x^2 + y^2 = 4x_0^2 + y_0^2$ for all t (direct differentiation: $8x\dot x + 2y\dot y = 8xy - 8xy = 0$). Then $\|\mathbf{x}(t)\|^2 = x^2 + y^2 \le 4x^2 + y^2 = 4x_0^2 + y_0^2 \le 4\|\mathbf{x}_0\|^2$. So given $\varepsilon > 0$ take $\delta = \varepsilon/2$: $\|\mathbf{x}_0\| < \delta \Rightarrow \|\mathbf{x}(t)\| \le 2\|\mathbf{x}_0\| < \varepsilon$ for all t: Liapunov stable. Not attracting: for $\mathbf{x}_0 \neq 0$, $4x^2 + y^2$ stays equal to $4x_0^2 + y_0^2 > 0$, so $\mathbf{x}(t) \not\to 0$, however small $\|\mathbf{x}_0\|$ is.

(b) The eigenvalues of $\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$ are $\pm\sqrt{2}$ with eigenvector $(\sqrt{2}, 1)$ for $+\sqrt{2}$. For any $\delta > 0$ the initial condition $\mathbf{x}_0 = \eta(\sqrt{2}, 1)$ with $\eta < \delta/\sqrt{3}$ has $\|\mathbf{x}_0\| < \delta$ but $\mathbf{x}(t) = \eta e^{\sqrt{2}t}(\sqrt{2}, 1)$ grows

without bound, so the solution leaves every ε-ball ($\varepsilon = 1$, say): not Liapunov stable, and it does not converge to 0: not attracting.

(c) $x = x_0$, $y = y_0 + x_0 t$. Take $\mathbf{x}_0 = (\eta, 0)$ with η arbitrarily small: $\|\mathbf{x}(t)\| \geq \eta t \to \infty$. So not Liapunov stable and not attracting.

(d) $x = x_0$, $y = y_0 e^{-t}$. Then $\|\mathbf{x}(t)\|^2 = x_0^2 + y_0^2 e^{-2t} \leq \|\mathbf{x}_0\|^2$ for $t \geq 0$, so $\delta = \varepsilon$ works: Liapunov stable. But $\mathbf{x}(t) \to (x_0, 0) \neq 0$ whenever $x_0 \neq 0$; for any δ the point $(\delta/2, 0)$ is within δ and does not tend to 0: not attracting.

(e) $x = x_0 e^{-t}$, $y = y_0 e^{-5t}$. For $t \geq 0$, $\|\mathbf{x}(t)\| \leq \|\mathbf{x}_0\|$, so $\delta = \varepsilon$ proves Liapunov stability; and $\mathbf{x}(t) \to 0$ for *every* initial condition, so any δ (e.g. $\delta = 1$) proves attraction. Hence asymptotically stable.

(f) $\mathbf{x}(t) = e^t \mathbf{x}_0$ grows exponentially for every $\mathbf{x}_0 \neq 0$: with $\varepsilon = 1$ and any $\delta > 0$ a point at distance $\delta/2$ leaves the unit ball, so it is not Liapunov stable, and no nonzero solution converges to 0: not attracting. ■

5.1.12 *Give a symmetry proof that the orbits of $\dot{x} = v$, $\dot{v} = -x$ are closed. (Hint: start on the v-axis at $(0, -v_0)$ and suppose the trajectory meets the x-axis at $(x, 0)$.)*

Solution. Two symmetries of the system will be used.

- *Reversibility:* if $(x(t), v(t))$ is a solution, so is $(x(-t), -v(-t))$. [Check: $X(t) = x(-t)$, $V(t) = -v(-t)$ gives $\dot{X} = -\dot{x}(-t) = -v(-t) = V$ and $\dot{V} = \dot{v}(-t) = -x(-t) = -X$.]
- *Linearity/odd symmetry:* if $(x(t), v(t))$ is a solution, so is $(-x(t), -v(t))$.

Start at $(0, -v_0)$ ($v_0 > 0$) at $t = 0$. There $\dot{x} = -v_0 < 0$, so the trajectory moves into the third quadrant ($x < 0$, $v < 0$). While there, $\dot{v} = -x > 0$. *It reaches the negative x-axis:* for $t \geq t_0 > 0$ we have $x(t) \leq x(t_0) < 0$, since x is decreasing, so $\dot{v} = -x \geq |x(t_0)| > 0$; hence v increases at least linearly and reaches 0 at some time T, at a point $(x_1, 0)$ with $x_1 < 0$.

Reflection about the first hit. Define $X(t) = x(2T - t)$, $V(t) = -v(2T - t)$. By reversibility (and time-translation invariance) this is a solution, and $X(T) = x_1$, $V(T) = -v(T) = 0$, the same state as $(x(T), v(T))$. By uniqueness of solutions, $X \equiv x$ and $V \equiv v$. In particular $x(2T) = x(0) = 0$ and $v(2T) = -v(0) = v_0$: at time $2T$ the trajectory is at $(0, v_0)$ on the positive v-axis.

Odd symmetry. The solution $(-x(t), -v(t))$ starts at $(0, v_0)$ at $t = 0$; by uniqueness the trajectory from $(0, v_0)$ at time $2T$ is $(-x(t - 2T), -v(t - 2T))$. So at time $3T$ it is at $(-x_1, 0)$ (on the positive x-axis), and, by the same reflection argument about that point, at time $4T$ it is at $(0, -v_0)$. Hence $(x(t + 4T), v(t + 4T)) = (x(t), v(t))$ for all t: the orbit is closed (with period $4T$). Since v_0 was arbitrary, every nonzero orbit is closed. ■

5.1.13 *Why do you think a "saddle point" is called by that name? What is the connection to real (horse) saddles?*

Solution. Consider the surface $z = x^2 - y^2$, which has the shape of a horse saddle. Along the x-direction it curves up (like the rider's seat from the middle toward the pommel and cantle) and along the y-direction it curves down (like the sides of the saddle toward the horse's flanks). If a marble (heavily damped) is placed on this surface, it rolls downhill following $\dot{\mathbf{x}} = -\nabla_{xy} z$, i.e. $\dot{x} = -2x$, $\dot{y} = +2y$: exactly the saddle-point flow of 5.1.9, with stable manifold the x-axis (the marble rolls back to the centre if it is displaced front-to-back) and unstable manifold the y-axis (it rolls off the sides if displaced sideways). The level curves of z are the hyperbolas of the phase portrait. So the fixed point is called a saddle because the flow there is that of a marble on a horse's saddle: stable in one direction, unstable in the other. ■

2 Classification of Linear Systems (5.2)

5.2.1 *$\dot{x} = 4x - y$, $\dot{y} = 2x + y$. (a) Write as $\dot{\mathbf{x}} = A\mathbf{x}$, show the characteristic polynomial is $\lambda^2 - 5\lambda + 6$, find eigenvalues and eigenvectors. (b) General solution. (c) Classify the origin. (d) Solve with $(x_0, y_0) = (3, 4)$.*

Solution. *(a)* $A = \begin{pmatrix} 4 & -1 \\ 2 & 1 \end{pmatrix}$, $\tau = 5$, $\Delta = 4 + 2 = 6$, so $\det(A - \lambda I) = \lambda^2 - \tau\lambda + \Delta = \lambda^2 - 5\lambda + 6 = (\lambda - 2)(\lambda - 3)$ and $\lambda_1 = 2$, $\lambda_2 = 3$. For $\lambda_1 = 2$: $A - 2I = \begin{pmatrix} 2 & -1 \\ 2 & -1 \end{pmatrix}$, $\mathbf{v}_1 = (1, 2)$. For $\lambda_2 = 3$: $A - 3I = \begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix}$, $\mathbf{v}_2 = (1, 1)$.

(b) $\mathbf{x}(t) = c_1 e^{2t} \begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_2 e^{3t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

(c) Both eigenvalues are positive ($\tau^2 - 4\Delta = 1 > 0$): an *unstable node*. The slow direction is $\mathbf{v}_1 = (1, 2)$ ($\lambda = 2$) and the fast direction is $\mathbf{v}_2 = (1, 1)$; trajectories leave the origin tangent to $(1, 2)$ and become parallel to $(1, 1)$ far away.

(d) $c_1 + c_2 = 3$ and $2c_1 + c_2 = 4$ give $c_1 = 1$, $c_2 = 2$:

$$x(t) = e^{2t} + 2e^{3t}, \qquad y(t) = 2e^{2t} + 2e^{3t}.$$

■

5.2.2 *(Complex eigenvalues) $\dot{x} = x - y$, $\dot{y} = x + y$. (a) Show $\lambda_{1,2} = 1 \pm i$ with eigenvectors $\mathbf{v}_1 = (i, 1)$, $\mathbf{v}_2 = (-i, 1)$. (b) Write the general solution in real form.*

Solution. *(a)* $A = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$: $\tau = 2$, $\Delta = 2$, $\lambda = \frac{1}{2}(2 \pm \sqrt{4 - 8}) = 1 \pm i$. For $\lambda_1 = 1 + i$: $(A - \lambda_1 I)\mathbf{v} = \begin{pmatrix} -i & -1 \\ 1 & -i \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0$ gives $v_2 = -iv_1$; multiplying by i, $\mathbf{v}_1 = (i, 1)$ (check: $-i \cdot i - 1 \cdot 1 = 1 - 1 = 0$). The conjugate eigenvalue $1 - i$ has the conjugate eigenvector $\mathbf{v}_2 = (-i, 1)$.

(b) $\mathbf{x} = c_1 e^{(1+i)t} \begin{pmatrix} i \\ 1 \end{pmatrix} + c_2 e^{(1-i)t} \begin{pmatrix} -i \\ 1 \end{pmatrix}$. Using $e^{it} = \cos t + i \sin t$,

$$e^{it} \begin{pmatrix} i \\ 1 \end{pmatrix} = \begin{pmatrix} -\sin t + i \cos t \\ \cos t + i \sin t \end{pmatrix}, \qquad e^{-it} \begin{pmatrix} -i \\ 1 \end{pmatrix} = \begin{pmatrix} -\sin t - i \cos t \\ \cos t - i \sin t \end{pmatrix}.$$

Hence

$$x = e^t\big[-(c_1 + c_2)\sin t + i(c_1 - c_2)\cos t\big], \qquad y = e^t\big[(c_1 + c_2)\cos t + i(c_1 - c_2)\sin t\big].$$

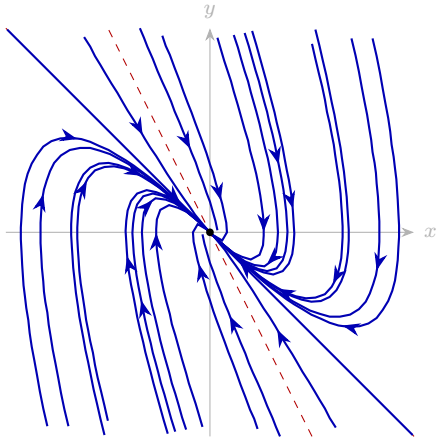
For real solutions $c_2 = \bar{c}_1$, so $C_1 := c_1 + c_2$ and $C_2 := i(c_1 - c_2)$ are real, and

$$\boxed{x(t) = e^t\big(C_2 \cos t - C_1 \sin t\big), \qquad y(t) = e^t\big(C_1 \cos t + C_2 \sin t\big)}$$

with $C_2 = x_0$, $C_1 = y_0$ (so $x = e^t(x_0 \cos t - y_0 \sin t)$, $y = e^t(y_0 \cos t + x_0 \sin t)$). (Check: $\dot{x} = e^t[(C_2 - C_1)\cos t - (C_1 + C_2)\sin t] = x - y$.) This is an unstable spiral, counterclockwise, with $r \propto e^t$ and angular speed 1 (indeed $\dot{r} = r$, $\dot{\theta} = 1$). ■

5.2.3 *$\dot{x} = y$, $\dot{y} = -2x - 3y$.*

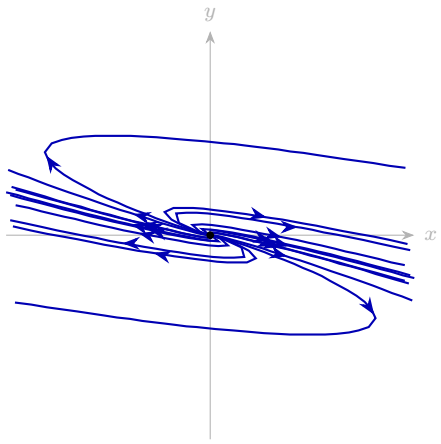
Solution. $A = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}$: $\tau = -3$, $\Delta = 2$, $\tau^2 - 4\Delta = 1 > 0$. Eigenvalues $\lambda = -1, -2$. For $\lambda = -1$: $A + I = \begin{pmatrix} 1 & 1 \\ -2 & -2 \end{pmatrix}$, $\mathbf{v} = (1, -1)$. For $\lambda = -2$: $A + 2I = \begin{pmatrix} 2 & 1 \\ -2 & -1 \end{pmatrix}$, $\mathbf{v} = (1, -2)$. The origin is a *stable node*. The slow eigendirection ($\lambda = -1$) is the line $y = -x$; the fast one is $y = -2x$. Trajectories approach the origin tangent to $y = -x$.



■

5.2.4 $\dot{x} = 5x + 10y$, $\dot{y} = -x - y$.

Solution. $A = \begin{pmatrix} 5 & 10 \\ -1 & -1 \end{pmatrix}$: $\tau = 4$, $\Delta = -5 + 10 = 5$, $\tau^2 - 4\Delta = 16 - 20 = -4 < 0$, eigenvalues $\lambda = 2 \pm i$. Since $\tau > 0$ the origin is an *unstable spiral*. At $(1, 0)$ the velocity is $(5, -1)$ (downward), so the rotation is clockwise. No real eigenvectors.

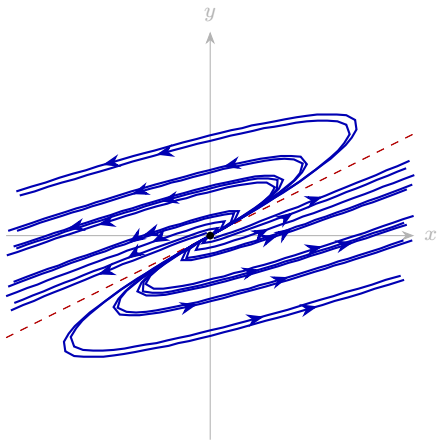


■

5.2.5 $\dot{x} = 3x - 4y$, $\dot{y} = x - y$.

Solution. $A = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}$: $\tau = 2$, $\Delta = -3 + 4 = 1$, $\tau^2 - 4\Delta = 0$: repeated eigenvalue $\lambda = 1$. $A - I = \begin{pmatrix} 2 & -4 \\ 1 & -2 \end{pmatrix} \neq 0$, so there is only one independent eigenvector, $\mathbf{v} = (2, 1)$ (the line $y = x/2$).

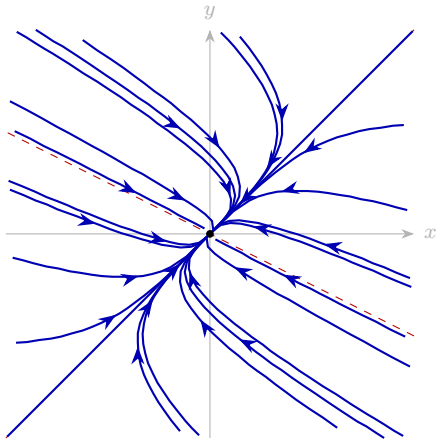
The origin is a *degenerate unstable node*. The angular velocity is $\dot\theta = (x\dot y - y\dot x)/r^2 = (x-2y)^2/r^2 \geq 0$, so trajectories turn counterclockwise, leave the origin tangent to $y = x/2$, and also approach the direction of $(2,1)$ for large t, bending around in a "S".



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5.2.6 $\dot x = -3x + 2y$, $\dot y = x - 2y$.

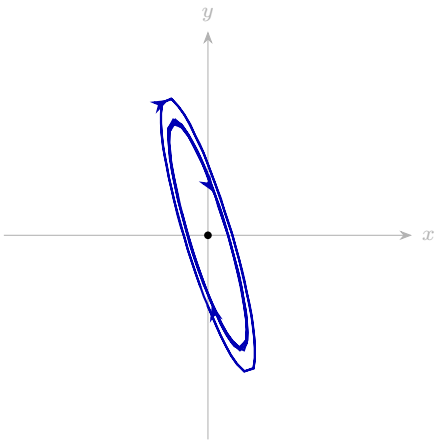
Solution. $A = \begin{pmatrix} -3 & 2 \\ 1 & -2 \end{pmatrix}$: $\tau = -5$, $\Delta = 6 - 2 = 4$, $\tau^2 - 4\Delta = 9$: $\lambda = \frac{1}{2}(-5 \pm 3) = -1, -4$. For $\lambda = -1$: $A + I = \begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix}$, $\mathbf{v} = (1,1)$. For $\lambda = -4$: $A + 4I = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}$, $\mathbf{v} = (2,-1)$. The origin is a *stable node*; the slow direction is $y = x$ ($\lambda = -1$), the fast direction is $y = -x/2$ ($\lambda = -4$). Trajectories enter the origin tangent to $y = x$.



■

5.2.7 $\dot x = 5x + 2y$, $\dot y = -17x - 5y$.

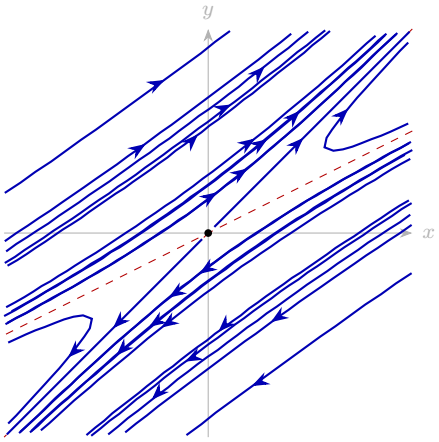
Solution. $A = \begin{pmatrix} 5 & 2 \\ -17 & -5 \end{pmatrix}$: $\tau = 0$, $\Delta = -25 + 34 = 9 > 0$, so $\lambda = \pm 3i$: a *center*. At $(1,0)$ the velocity is $(5,-17)$ (downward), so the orbits are traversed clockwise. They are tilted ellipses (the quadratic form $17x^2 + 10xy + 2y^2$ is conserved: $\frac{\mathrm{d}}{\mathrm{d}t}[17x^2 + 10xy + 2y^2] = (34x + 10y)(5x + 2y) + (10x + 4y)(-17x - 5y) = 0$), with period $2\pi/3$.



■

5.2.8 $\dot{x} = -3x + 4y$, $\dot{y} = -2x + 3y$.

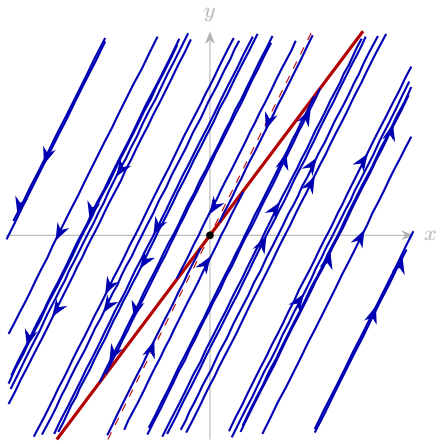
Solution. $A = \begin{pmatrix} -3 & 4 \\ -2 & 3 \end{pmatrix}$: $\tau = 0$, $\Delta = -9 + 8 = -1 < 0$: a *saddle*, $\lambda = \pm 1$. For $\lambda = 1$: $A - I = \begin{pmatrix} -4 & 4 \\ -2 & 2 \end{pmatrix}$, $\mathbf{v} = (1, 1)$. For $\lambda = -1$: $A + I = \begin{pmatrix} -2 & 4 \\ -2 & 4 \end{pmatrix}$, $\mathbf{v} = (2, 1)$. So the unstable manifold is $y = x$ and the stable manifold is $y = x/2$.



■

5.2.9 $\dot{x} = 4x - 3y$, $\dot{y} = 8x - 6y$.

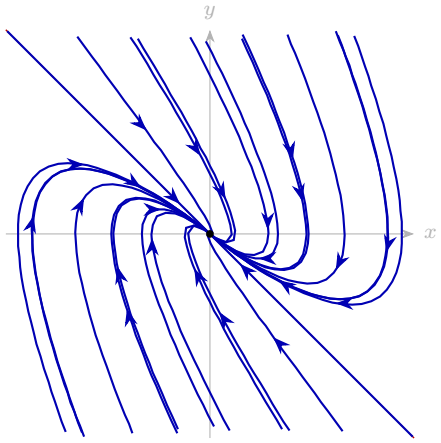
Solution. $A = \begin{pmatrix} 4 & -3 \\ 8 & -6 \end{pmatrix}$: $\tau = -2$, $\Delta = -24 + 24 = 0$: eigenvalues $\lambda = 0$ and $\lambda = -2$. For $\lambda = 0$: $4x - 3y = 0$, $\mathbf{v} = (3, 4)$; for $\lambda = -2$: $A + 2I = \begin{pmatrix} 6 & -3 \\ 8 & -4 \end{pmatrix}$, $\mathbf{v} = (1, 2)$. Since $\Delta = 0$ the fixed points are not isolated: the whole line $y = \frac{4}{3}x$ consists of fixed points. The general solution is $\mathbf{x} = c_1(3, 4) + c_2 e^{-2t}(1, 2)$: every other trajectory is a straight line parallel to $(1, 2)$, moving toward the line of fixed points (stable, non-isolated).



■

5.2.10 $\dot{x} = y$, $\dot{y} = -x - 2y$.

Solution. $A = \begin{pmatrix} 0 & 1 \\ -1 & -2 \end{pmatrix}$: $\tau = -2$, $\Delta = 1$, $\tau^2 - 4\Delta = 0$: $\lambda = -1$ (repeated). $A + I = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix} \neq 0$: a single eigenvector $(1, -1)$ (the line $y = -x$). The origin is a *degenerate stable node*. The angular velocity is $\dot{\theta} = (x\dot{y} - y\dot{x})/r^2 = -(x+y)^2/r^2 \leq 0$: clockwise turning. Trajectories enter the origin tangent to $y = -x$.



■

5.2.11 *Show that* $A = \begin{pmatrix} \lambda & b \\ 0 & \lambda \end{pmatrix}$, $b \neq 0$, *has only a one-dimensional eigenspace for* λ. *Solve* $\dot{\mathbf{x}} = A\mathbf{x}$ *and sketch the phase portrait.*

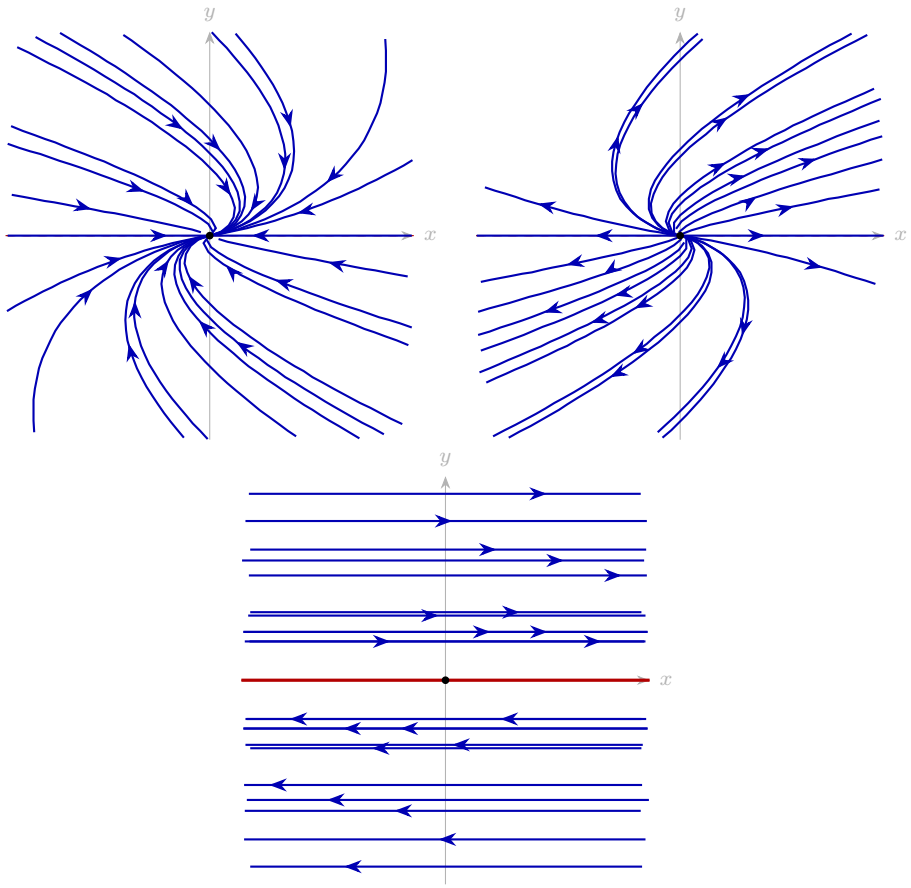
Solution. $A - \lambda I = \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix}$, whose null space is $\{\mathbf{v} : bv_2 = 0\} = \text{span}\{(1, 0)\}$ (because $b \neq 0$): one-dimensional. Solving the system from the bottom: $\dot{y} = \lambda y \Rightarrow y = y_0 e^{\lambda t}$; then $\dot{x} = \lambda x + by_0 e^{\lambda t}$ gives

$$x(t) = e^{\lambda t}(x_0 + by_0 t), \qquad y(t) = y_0 e^{\lambda t}.$$

- $\lambda < 0$: a degenerate stable node; all trajectories approach the origin tangent to the x-axis.

- $\lambda > 0$: a degenerate unstable node (trajectories leave tangent to the x-axis).
- $\lambda = 0$: $x = x_0 + by_0t$, $y = y_0$: a shear flow; every point of the x-axis is fixed and the other trajectories are horizontal lines, moving right in the upper half-plane and left in the lower half-plane if $b > 0$ (reversed if $b < 0$).

The figures show $\lambda = -1, b = 1$ (left), $\lambda = 1, b = 1$ (middle), $\lambda = 0, b = 1$ (right).



■

5.2.12 *(LRC circuit)* $L\ddot{I} + R\dot{I} + I/C = 0$, $L, C > 0$, $R \geq 0$. *(a) Rewrite as a 2D linear system. (b) Show the origin is asymptotically stable if $R > 0$ and neutrally stable if $R = 0$. (c) Classify the origin depending on the sign of $R^2C - 4L$ and sketch.*

Solution. *(a)* Let $x = I$, $y = \dot{I}$. Then

$$\dot{x} = y, \qquad \dot{y} = -\frac{1}{LC}x - \frac{R}{L}y, \qquad A = \begin{pmatrix} 0 & 1 \\ -1/(LC) & -R/L \end{pmatrix}.$$

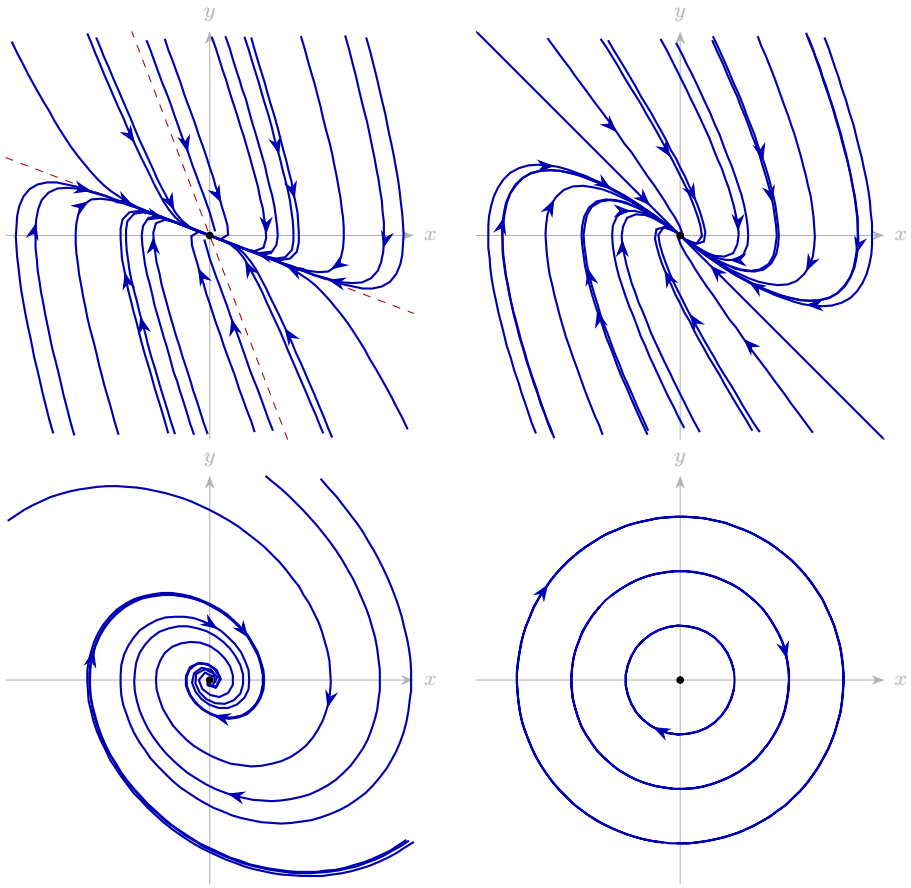
(b) $\tau = -R/L \leq 0$ and $\Delta = 1/(LC) > 0$. If $R > 0$: $\tau < 0$ and $\Delta > 0$, so both eigenvalues have negative real part and the origin is asymptotically stable (a stable node or spiral). If $R = 0$: $\tau = 0$, $\lambda = \pm i/\sqrt{LC}$, a center: neutrally (Liapunov) stable but not attracting.

(c) $\tau^2 - 4\Delta = \dfrac{R^2}{L^2} - \dfrac{4}{LC} = \dfrac{R^2C - 4L}{L^2C}$, which has the sign of $R^2C - 4L$:

- $R^2C - 4L > 0$ (large resistance): two negative real eigenvalues, a *stable node*: the current decays without oscillating (overdamped).

- $R^2C - 4L = 0$: $\lambda = -R/(2L)$ repeated, a *degenerate stable node* (critically damped).
- $R^2C - 4L < 0$ (small resistance): complex eigenvalues $-\frac{R}{2L} \pm i\omega$, $\omega = \sqrt{1/(LC) - R^2/(4L^2)}$, a *stable spiral*: decaying oscillations (underdamped).
- (For $R = 0$ the discriminant is $-4L < 0$ and the spiral becomes a center: undamped oscillation at frequency $1/\sqrt{LC}$.)

Sketches (for concreteness $L = C = 1$ and $R = 3$, 2, 0.5, 0, in the $(I, \dot{I})$-plane):



■

5.2.13 *(Damped harmonic oscillator)* $m\ddot{x} + b\dot{x} + kx = 0$, $b > 0$. *(a) Two-dimensional form. (b) Classify the origin in all cases. (c) Relation to overdamped, critically damped, underdamped.*

Solution. *(a)* With $v = \dot{x}$: $\dot{x} = v$, $\dot{v} = -\frac{k}{m}x - \frac{b}{m}v$, so $A = \begin{pmatrix} 0 & 1 \\ -k/m & -b/m \end{pmatrix}$.

(b) $\tau = -b/m < 0$, $\Delta = k/m > 0$ and $\tau^2 - 4\Delta = \dfrac{b^2 - 4mk}{m^2}$. Hence:

- $b^2 > 4mk$: stable node (two negative real eigenvalues $\lambda_\pm = \dfrac{-b \pm \sqrt{b^2 - 4mk}}{2m}$); the phase portrait is as in the first sketch of 5.2.12, trajectories enter the origin tangent to the slow eigendirection $v = \lambda_+ x$.
- $b^2 = 4mk$: degenerate stable node (repeated $\lambda = -b/2m$), second sketch of 5.2.12.

- $b^2 < 4mk$: stable spiral, $\lambda = -\dfrac{b}{2m} \pm i\omega$, $\omega = \sqrt{k/m - b^2/(4m^2)}$ (third sketch of 5.2.12).

(As $b \to 0^+$ the spiral tightens to a center, the undamped oscillator.) The origin is always asymptotically stable for $b > 0$.

(c) $b^2 > 4mk$ is *overdamped* (the mass creeps back to equilibrium without oscillating; it crosses $x = 0$ at most once); $b^2 = 4mk$ is *critically damped* (the fastest return without oscillation); $b^2 < 4mk$ is *underdamped* (the mass oscillates with exponentially decaying amplitude $e^{-bt/2m}$ and frequency ω). The node/degenerate node/spiral classification of the origin is exactly this trichotomy. ■

5.2.14 *(A project about random systems) Pick* $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ *with* a, b, c, d *independent and uniform on* $[-1, 1]$. *Find the probabilities of all the different kinds of fixed points. Check by Monte Carlo. Are the answers the same for a normal distribution?*

Solution. Symmetries. (i) $\Delta = ad - bc < 0$ iff $ad < bc$. The variables ad and bc are independent, identically distributed and symmetric about 0, so $P(ad < bc) = \frac{1}{2}$ (continuous distribution). Hence $P(\text{saddle}) = \frac{1}{2}$, for the uniform *and* the normal distribution. (ii) $A \mapsto -A$ preserves the distribution and swaps $\tau \mapsto -\tau$, $\Delta \mapsto \Delta$, so stable and unstable nodes are equally likely, as are stable and unstable spirals. (iii) Centers ($\tau = 0$), degenerate nodes/stars ($\tau^2 = 4\Delta$), and lines of fixed points ($\Delta = 0$) occur on sets of measure zero, so their probability is 0. Since complex eigenvalues imply $\Delta = |\lambda|^2 > 0$, we get

$$P(\text{nodes}) = \tfrac{1}{2} - P(\text{spirals}), \qquad P(\text{spirals}) = P(\text{complex eigenvalues}) = P\big((a-d)^2 + 4bc < 0\big).$$

Uniform case. Let $u = a - d$; it has the triangular density $f(u) = (2 - |u|)/4$ on $[-2, 2]$. For independent X, Y uniform on $(0, 1)$, $P(XY > z) = 1 - z + z\ln z$ for $0 \le z \le 1$, so $P(bc < -z) = \frac{1}{2}(1 - z + z\ln z)$. Complex eigenvalues require $bc < -u^2/4$, so with $z = u^2/4$:

$$P(\text{complex}) = \int_{-2}^{2} \frac{2 - |u|}{4} \cdot \frac{1}{2}\Big(1 - z + z\ln z\Big)\mathrm{d}u = \frac{23}{72} \approx 0.3194.$$

Therefore (uniform):

type	exact	Monte Carlo (10^8 matrices)
saddle	$\frac{1}{2}$	0.4999_8
stable node	$\frac{13}{144} \approx 0.0903$	0.0903
unstable node	$\frac{13}{144} \approx 0.0903$	0.0903
stable spiral	$\frac{23}{144} \approx 0.1597$	0.1597
unstable spiral	$\frac{23}{144} \approx 0.1597$	0.1598
center, star, degenerate node, line of fixed points	0	–

Normal case (a, b, c, d i.i.d. $N(0,1)$). Write $a - d = \sqrt{2}\,p$, $b + c = \sqrt{2}\,s$, $b - c = \sqrt{2}\,t$ with p, s, t i.i.d. $N(0,1)$; then $4bc = (b+c)^2 - (b-c)^2 = 2s^2 - 2t^2$, and complex eigenvalues mean $2p^2 + 2s^2 - 2t^2 < 0$, i.e. $p^2 + s^2 < t^2$. Given t, $p^2 + s^2$ is exponential with mean 2, so $P(p^2 + s^2 < t^2 \mid t) = 1 - e^{-t^2/2}$, and

$$P(\text{complex}) = 1 - \mathbb{E}\, e^{-t^2/2} = 1 - \frac{1}{\sqrt{2}} \approx 0.2929.$$

(Normal case:)

type	exact	Monte Carlo (10^8 matrices)
saddle	$\frac{1}{2}$	0.4999
stable / unstable node (each)	$\frac{1}{2}\big(\frac{1}{\sqrt{2}} - \frac{1}{2}\big) \approx 0.1036$	0.1036
stable / unstable spiral (each)	$\frac{1}{2}\big(1 - \frac{1}{\sqrt{2}}\big) \approx 0.1464$	0.1465

Comparison. The saddle probability ($\frac{1}{2}$) and the symmetry statements are the same for both distributions, but the split between nodes and spirals differs: with uniform entries nodes are less likely (0.1806 total) and spirals more likely (0.3194) than with Gaussian entries (0.2071 and 0.2929). ■

3 Love Affairs (5.3)

5.3.1 *(Name-calling) Suggest names for the four romantic styles determined by the signs of a and b in $\dot{R} = aR + bJ$.*

Solution. Here a measures how Romeo responds to his *own* feelings and b how he responds to Juliet's.

signs	name	description
$a > 0,\ b > 0$	*eager beaver*	excited by her love and spurred on by his own feelings
$a < 0,\ b > 0$	*cautious lover*	responds to her love but backs off when his own feelings grow strong
$a > 0,\ b < 0$	*narcissistic nerd*	feeds on his own feelings but is turned off by hers
$a < 0,\ b < 0$	*hermit*	recoils from both his own feelings and hers

(Other names are of course possible; the first two are from the text.) ■

5.3.2 *Consider $\dot{R} = J$, $\dot{J} = -R + J$. (a) Characterize the romantic styles of Romeo and Juliet. (b) Classify the origin and interpret. (c) Sketch $R(t), J(t)$ for $R(0) = 1$, $J(0) = 0$.*

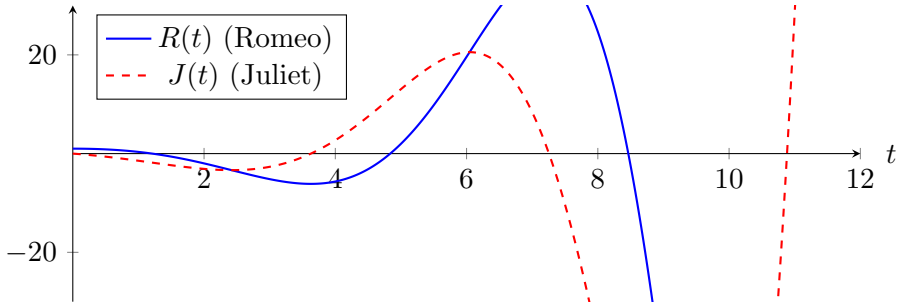
Solution. *(a)* Romeo: $a = 0$, $b = 1$: he ignores his own feelings and simply follows Juliet's (pure responder). Juliet: $c = -1$ (turned off by Romeo's love), $d = 1$ (spurred by her own feelings): a *narcissistic nerd* (she grows in love with herself and retreats from his advances).

(b) $A = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}$: $\tau = 1$, $\Delta = 1$, $\tau^2 - 4\Delta = -3 < 0$: eigenvalues $\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$, an *unstable spiral*. The feelings oscillate between love and hate with period $4\pi/\sqrt{3} \approx 7.26$ and the amplitude grows like $e^{t/2}$: an ever more stormy affair, with wilder and wilder swings (in this linear model without bound). The only way to indifference is the unattainable $R = J = 0$.

(c) $\dot{R} = J$ gives $\ddot{R} = \dot{J} = -R + \dot{R}$, i.e. $\ddot{R} - \dot{R} + R = 0$, so $R = e^{t/2}\big(A\cos\frac{\sqrt{3}}{2}t + B\sin\frac{\sqrt{3}}{2}t\big)$. $R(0) = 1$ gives $A = 1$; $\dot{R}(0) = J(0) = 0$ gives $\frac{A}{2} + \frac{\sqrt{3}}{2}B = 0$, $B = -1/\sqrt{3}$. Therefore

$$R(t) = e^{t/2}\Big(\cos\tfrac{\sqrt{3}}{2}t - \tfrac{1}{\sqrt{3}}\sin\tfrac{\sqrt{3}}{2}t\Big), \qquad J(t) = \dot{R} = -\tfrac{2}{\sqrt{3}}\,e^{t/2}\sin\tfrac{\sqrt{3}}{2}t.$$

(Check: $\dot{J}(0) = -1 = -R(0) + J(0)$.) Juliet turns cold immediately ($J < 0$ for $0 < t < 2\pi/\sqrt{3} \approx 3.63$), while Romeo's love decays, crossing zero at $t = 2\pi/3 \approx 2.09$; then both change sign repeatedly with growing amplitude.



■

5.3.3 *(Out of touch with their own feelings)* $\dot R = aJ$, $\dot J = bR$. *What happens?*

Solution. $A = \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix}$: $\tau = 0$, $\Delta = -ab$, $\lambda^2 = ab$.

- $ab > 0$ (both react to the other in the same sense): $\lambda = \pm\sqrt{ab}$, a *saddle*. The unstable eigenvector satisfies $J = \sqrt{ab}\,R/a$. If $a, b > 0$ this is $J = \sqrt{b/a}\,R$: Romeo and Juliet end up with feelings of the same sign that escalate exponentially, mutual love if $R + \sqrt{a/b}J > 0$ initially, mutual hate if negative (plus a one-parameter family that decays to indifference, the stable manifold $J = -\sqrt{b/a}R$). If $a, b < 0$ the unstable direction has $J = -\sqrt{b/a}R$: opposite signs, i.e. one loves and the other hates, with escalating intensity (unrequited love).
- $ab < 0$ (one chases, the other runs away): $\lambda = \pm i\sqrt{-ab}$, a *center*: perpetual oscillation, love and hate alternating forever with constant amplitude and period $2\pi/\sqrt{-ab}$. The orbits are ellipses $bR^2 - aJ^2 = \text{const}$ (indeed $\frac{\mathrm{d}}{\mathrm{d}t}(bR^2 - aJ^2) = 2baRJ - 2abJR = 0$). Romeo and Juliet are "out of phase" (he loves her when she hates him, etc.).
- $ab = 0$: degenerate (e.g. $a = 0, b \neq 0$: $R = R_0$ constant, $J = J_0 + bR_0 t$ grows linearly).

■

5.3.4 *(Fire and water)* $\dot R = aR + bJ$, $\dot J = -bR - aJ$. *Do opposites attract?*

Solution. $A = \begin{pmatrix} a & b \\ -b & -a \end{pmatrix}$: $\tau = 0$, $\Delta = -a^2 + b^2$, $\lambda^2 = a^2 - b^2$.

- $b^2 > a^2$: $\Delta > 0$, $\lambda = \pm i\sqrt{b^2 - a^2}$: a *center*. Their feelings oscillate forever, with period $2\pi/\sqrt{b^2 - a^2}$, never settling: an endless cycle of love and hate.
- $b^2 < a^2$: $\Delta < 0$, a *saddle* with $\lambda = \pm\sqrt{a^2 - b^2}$: for almost every initial condition the feelings grow exponentially (along the unstable eigenvector $J = (\mu - a)R/b$, $\mu = \sqrt{a^2 - b^2}$). Only on the stable manifold do they decay to indifference.
- $b^2 = a^2$ (and $A \neq 0$): $A^2 = 0$, $\Delta = \tau = 0$: linear growth, $\mathbf{x} = \mathbf{x}_0 + tA\mathbf{x}_0$.

So opposites do *not* attract in the sense of settling into a stable, mutually satisfied relationship: either the affair cycles forever (when the responsiveness $|b|$ beats the self-reaction $|a|$) or it blows up (when $|a| > |b|$). It is never asymptotically stable ($\tau = 0$). ■

5.3.5 *(Peas in a pod)* $\dot R = aR + bJ$, $\dot J = bR + aJ$. *Should they expect boredom or bliss?*

Solution. This is Example 5.3.1 with arbitrary signs. Let $u = R + J$ and $v = R - J$: then $\dot u = (a + b)u$ and $\dot v = (a - b)v$, so $u = u_0 e^{(a+b)t}$, $v = v_0 e^{(a-b)t}$ (eigenvalues $a \pm b$, eigenvectors $(1, 1)$ and $(1, -1)$). The outcome:

- $a < -|b|$ (both eigenvalues negative; stable node): $R, J \to 0$: *boredom* (the affair fizzles out to indifference). This is the cautious regime: each is too wary of their own feelings.
- $a > |b|$ (both positive; unstable node): feelings run away exponentially; since $u = R+J$ grows at the faster rate when $b > 0$, eventually $R \approx J$: mutual love (*bliss*) if $R_0 + J_0 > 0$, mutual hate if < 0 (for $b < 0$ the v mode dominates and $R \approx -J$).
- $|a| < |b|$ (saddle): if $b > 0$, the unstable mode is u, so $R \approx J$ grows without bound: *bliss* if $R_0 + J_0 > 0$, *mutual misery* if < 0; and indifference only if $R_0 + J_0 = 0$. If $b < 0$ the unstable mode is $v = R - J$ and they end up with opposite feelings (one loves, the other hates: unrequited love).

So the fate of two identical lovers depends on the signs of $a \pm b$ and on the initial conditions; bliss is possible, specifically when $b > 0$, $a + b > 0$ and $R_0 + J_0 > 0$, but so are boredom and mutual hatred. ■

5.3.6 *(Romeo the robot) $\dot{R} = 0$, $\dot{J} = aR + bJ$. Does Juliet end up loving him or hating him?*

Solution. $R = R_0$ is constant. Then $\dot{J} = bJ + aR_0$ is a linear first-order equation.

- $b < 0$ (Juliet is cautious about her own feelings): $J(t) = -\dfrac{aR_0}{b} + \Big(J_0 + \dfrac{aR_0}{b}\Big)e^{bt} \to J^* = \dfrac{aR_0}{|b|}$. She ends up loving him if $aR_0 > 0$ (e.g. he loves her, $R_0 > 0$, and she is drawn by his love, $a > 0$), and hating him if $aR_0 < 0$. The final level is proportional to Romeo's constant feeling.
- $b > 0$ (Juliet is an eager beaver about her own feelings): $J(t) = -\dfrac{aR_0}{b} + \Big(J_0 + \dfrac{aR_0}{b}\Big)e^{bt}$ blows up exponentially; she ends up loving him (infinitely) if $J_0 + aR_0/b > 0$ and hating him if $J_0 + aR_0/b < 0$. Her initial feeling matters, not just Romeo's.
- $b = 0$: $J = J_0 + aR_0 t$: linear drift, so she eventually loves him if $aR_0 > 0$ and hates him if $aR_0 < 0$.

(If $R_0 = 0$ the robot is indifferent: $J = J_0 e^{bt}$, which decays for $b < 0$ and grows in the direction of J_0 for $b > 0$.) ■